AROVIA.AI — COMPLETE CLINICAL CHAT & DISEASE DOSSIER

All 18 Personas • Multi-Turn Consultations • Pain Perception Models • Bayesian Inference

Engine: 100% Deterministic Local Engine (Zero External API Dependency) | **Date:** October 2026 | **Target:** Physician Audit

CLINICAL OVERVIEW FOR ATTENDING PHYSICIANS:
This dossier provides a 360-degree clinical audit of Arovia's clinical engine across **all 18 patient personas** and **all core disease presentations**. Unlike generic LLM wrappers, this engine operates **100% on the local deterministic codebase** (PersonaEngine, MCMCEngine, and SymptomQuestionSchemas), using Bayesian prior odds updates, multi-system organ strain models, and Beers/FDA safety filters. Crucially, it captures how patients in real life describe their symptoms: stoic grandfathers who minimize heart attacks as 'gas', anxious developers tracking Apple Watch biometrics, truckers self-medicating on NSAIDs until stomach ulcers bleed, and pregnant mothers safeguarding their unborn child.

Master Persona Demographics & Baseline Vitals (All 18 Profiles)

ID	Patient Name	Age/Sex	Occupation	BMI / BP	Chronic Illnesses	Frailty	Polypharmacy
#1	**Ramesh Sharma**	68y M	Retired Bank Clerk	29.8 | 148/92	Type 2 Diabetes Mellitus (15 yrs), Essential Hypertension (18 yrs), Stage 3a Chronic Kidney Disease (eGFR 48 mL/min), Diabetic Retinopathy (Mild)	6/10	HIGH
#2	**Aanya Sen**	23y F	Junior Software Engineer	19.4 | 106/70	Polycystic Ovary Syndrome (PCOS, on Combined Oral Contraceptive therapy), Chronic Psychogenic Insomnia, Generalized Anxiety Disorder	2/10	LOW
#3	**Gurpreet Singh**	42y M	Interstate Long-Haul Truck Driver	34.6 | 154/98	Severe Obstructive Sleep Apnea (OSA, untreated), Non-Alcoholic Fatty Liver Disease (NAFLD, Grade 2), Chronic Lumbar Spondylosis & Sciatica	4/10	LOW
#4	**Meera Nambiar ■**	32y F	High School Biology Teacher	25.6 | 112/74	Gestational Anemia (Hb 9.8 g/dL), Mild Gestational Thrombocytopenia (Platelets 135k), Morning Sickness / Reflux	2/10	LOW
#5	**Vikramaditya 'Vik' Rathore**	19y M	Collegiate 400m Sprinter & Kinesiology Student	22.7 | 108/66	Exercise-Induced Bronchoconstriction (EIB / Asthma), History of recurrent Right Hamstring Strain	0/10	NONE
#6	**Devaki Patel**	74y F	Homemaker & Grandmother	18.7 | 134/82	Severe Osteoporosis (T-score -3.4, prior Colles' wrist fracture), Bilateral Knee Osteoarthritis (Kellgren Grade 3), Early Vascular Cognitive Impairment, Chronic Postural Instability	7/10	MODERATE
#7	**Sunita Bannerjee**	49y F	VP of Corporate Risk & Regulatory Compliance	26.4 | 138/88	Hashimoto's Autoimmune Hypothyroidism, Dyslipidemia (Elevated LDL 154 mg/dL), Chronic Occupational Burnout Syndrome	3/10	LOW
#8	**Karthik 'Koko' Roy**	28y M	Nightclub Mixologist & Audio Sound Producer	23.7 | 126/82	Chronic Gastroesophageal Reflux Disease (GERD with Grade B Esophagitis), Bilateral Noise-Induced Tinnitus, Social Anxiety Disorder	2/10	MODERATE
#9	**Dr. Farooq Qureshi**	58y M	Senior Cardiothoracic Surgeon	26.8 | 142/90	Coronary Artery Disease (s/p Drug-Eluting Stent to LAD 4 yrs ago), Asymptomatic Hyperuricemia & Mild Gout, Bilateral Lower Extremity Varicose Veins	4/10	MODERATE
#10	**Priyanka Rao**	36y F	Permaculture Farmer & Ashtanga Yoga Acharya	19.5 | 102/66	Irritable Bowel Syndrome (IBS-Diarrhea predominant, stress/fodmap triggered), Contact Allergic Phytodermatitis (Parthenium plant resins)	1/10	NONE
#11	**Anand Joshi**	54y M	Senior Chemical Dye & Textile Mill Foreman	22.8 | 132/86	Occupational Chronic Bronchitis / Byssinosis, Mild Diabetic Neuropathy (burning feet), History of Peptic Duodenal Ulcer (bleeding 6 yrs ago)	5/10	MODERATE
#12	**Zoya Sheikh**	16y F	High School Science Student & Competitive Swimmer	19.3 | 104/68	Severe Extrinsic Atopic Dermatitis (eczema in cubital & popliteal fossae), Seasonal Allergic Rhinitis, Mild Exercise-Induced Rhinorrhea	0/10	LOW
#13	**Harbans Lal**	81y M	Assisted Living Resident / Retired Railway Mechanical Engineer	20.6 | 116/68	Non-Valvular Atrial Fibrillation (CHADS-VASc 5, on Apixaban), Congestive Heart Failure (NYHA Class III, Ejection Fraction 32%), Mixed Vascular & Alzheimer's Dementia, Stage 4 Chronic Kidney Disease (eGFR 22 mL/min), Benign Prostatic Hyperplasia with chronic catheterization history, Severe Sarcopenia & High Aspiration Risk	9/10	CRITICAL
#14	**Radhika Menon**	41y F	Clinical Psychologist & Psychotherapist	31.2 | 136/86	Fibromyalgia (14/18 bilateral tender points), Major Depressive Disorder (in clinical remission on maintenance), Chronic Migraine without Aura	4/10	MODERATE
#15	**Manish Chawla**	47y M	Commercial Real Estate Developer & Broker	31.7 | 156/96	Severe Recurrent Tophaceous Gout (Serum Uric Acid 9.8 mg/dL), Metabolic Syndrome with Central Adiposity, Grade 2 Hepatic Steatosis (Fatty Liver), Stage 1 Essential Hypertension	4/10	MODERATE
#16	**Tenzin Norbu**	37y M	Himalayan Expedition High-Altitude Mountaineering Guide	22.8 | 114/72	Past High Altitude Pulmonary Edema (HAPE episode 4 yrs ago at 6,200m), Cold Urticaria (erythema upon sub-zero exposure), Actinic Solar Cheilitis	1/10	LOW
#17	**Shalini Kulkarni**	63y F	Classical Bharatanatyam Guru & Dance Academy Director	22.9 | 128/80	Bilateral Severe Ankle Subtalar Osteoarthritis, Medial Meniscal Degeneration of Right Knee, Mild Lumbar Osteopenia	3/10	LOW
#18	**Deepak Verma**	34y M	High-Frequency Financial Derivatives Trader	21.5 | 144/92	Severe Chronic Tension-Type Headaches, Severe Computer Vision Syndrome & Evaporative Dry Eye, Functional Orthostatic Tachycardia / Adrenaline-Induced Palpitations, Labile Stress Hypertension	2/10	LOW

1. Cardiovascular: Acute Crushing Chest Pain & Arm Radiation

Clinical Domain: Cardiology / Coronary Ischemia vs. Costochondritis vs. GERD | **Complaint:** Severe retrosternal pressure, feels like an elephant sitting on chest, radiating to left shoulder and jaw

A. Complete 18-Persona Risk & Organ Strain Comparison Matrix

Persona	Triage	Top Calculated Diagnosis	Posterior	Prior Shift (α)	Key Organ Strain	Primary DDI / Safety Alert
#1 Ramesh Sharma 68y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**73.3%**	5.72x	Cardio: 100% Renal: 90%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#2 Aanya Sen 23y F	URGENT (RULE-OUT)	**Costochondritis / Musculoskeletal Chest Wall Strain**	**64.4%**	2.8x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#3 Gurpreet Singh 42y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**53.6%**	2.52x	Cardio: 90% Renal: 55%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#4 Meera Nambiar 32y F	URGENT (RULE-OUT)	**Gastroesophageal Reflux / Esophageal Spasm**	**59.4%**	2.4x	Cardio: 90% Renal: 40%	■ **PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimest...**
#5 Vikramaditya 'Vik' Rathore 19y M	URGENT (RULE-OUT)	**Costochondritis / Musculoskeletal Chest Wall Strain**	**64.4%**	2.8x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#6 Devaki Patel 74y F	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**50.4%**	2.6x	Cardio: 100% Renal: 55%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#7 Sunita Bannerjee 49y F	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**55.5%**	2.6x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#8 Karthik 'Koko' Roy 28y M	URGENT (RULE-OUT)	**Gastroesophageal Reflux / Esophageal Spasm**	**41.7%**	2.4x	Cardio: 90% Renal: 40%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#9 Dr. Farooq Qureshi 58y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**81.4%**	9.1x	Cardio: 100% Renal: 40%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#10 Priyanka Rao 36y F	URGENT (RULE-OUT)	**Gastroesophageal Reflux / Esophageal Spasm**	**40.9%**	0.9x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#11 Anand Joshi 54y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**55.5%**	2.6x	Cardio: 90% Renal: 40%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#12 Zoya Sheikh 16y F	URGENT (RULE-OUT)	**Costochondritis / Musculoskeletal Chest Wall Strain**	**64.4%**	2.8x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#13 Harbans Lal 81y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**50.4%**	2.6x	Cardio: 100% Renal: 90%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#14 Radhika Menon 41y F	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**40.2%**	1.4x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#15 Manish Chawla 47y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**63.6%**	3.64x	Cardio: 100% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#16 Tenzin Norbu 37y M	URGENT (RULE-OUT)	**Gastroesophageal Reflux / Esophageal Spasm**	**40.9%**	0.9x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#17 Shalini Kulkarni 63y F	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**55.5%**	2.6x	Cardio: 90% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#18 Deepak Verma 34y M	EMERGENCY	**Acute Coronary Syndrome (STEMI / NSTEMI)**	**45.2%**	1.8x	Cardio: 100% Renal: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.

B. Synthetic Clinical Simulation Transcripts (Multi-Turn Triage Benchmark)

PATIENT CONSULTATION: #1 RAMESH SHARMA (68y male) — Retired Bank Clerk

■ **Ramesh Sharma:** "Namaste beta... My daughter forced me to consult. It is probably just some gas from the evening samosa. But since 45 minutes there is a heavy stone-like pressure in the center of my chest, and my left shoulder feels slightly tired. Nothing serious, I already took some ajwain water."

■ **Arovia Health Assistant:** "Hello Mr. Sharma. Please stay completely still. In someone with 15 years of diabetes, chest heaviness and left shoulder fatigue must be treated as a potential heart event requiring emergency evaluation. A normal resting ECG alone cannot rule out acute coronary syndrome. Are you sweating, feeling lightheaded, or having trouble catching your breath?"

■ **Ramesh Sharma:** "Yes beta... My undershirt is soaked with cold sweat, which is strange because the fan is on. Breathing feels a bit shallow, and my head was spinning when I stood up from bed."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: Cold sweats with crushing retrosternal pressure in diabetes is an emergency sign of acute coronary ischemia. Do not eat, drink, or take any painkillers. Call an ambulance (112) immediately; do not drive yourself."

■ **Ramesh Sharma:** "Understood beta, my son is calling the 112 ambulance right now. Resting quietly until they arrive."

■ **CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Acute Coronary Syndrome (STEMI / NSTEMI) (Posterior: 73.3%, Multiplier: 5.72x)**
Safety / DDI Filter: ■ RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly contraindicated in Stage 3-5 CKD. High risk of hemodynamically mediated Acute Kidney Injury (AKI) and accelerating CKD progression via afferent arteriolar vasoconstriction.

PATIENT CONSULTATION: #2 AANYA SEN (23y female) — Junior Software Engineer

■ **Aanya Sen:** "Doctor omg please help me!! My Apple Watch literally just buzzed an alert that my resting HR hit 138 bpm! My chest feels so tight, like an iron band is clamping my ribs shut, and my left arm has pins-and-needles. Am I having a heart attack?! I'm breathing into a paper bag now to stop hyperventilating!"

■ **Arovia Health Assistant:** "■■ URGENT SAFETY INTERCEPT: Aanya, STOP breathing into a paper bag immediately! Rebreathing into a paper bag is medically dangerous—it causes progressive hypoxia and can mask serious cardiopulmonary emergencies. Take slow, unassisted breaths. Does the pain become sharply worse when you press directly on your ribs or take a deep breath?"

■ **Aanya Sen:** "Threw the bag away, breathing on my own. Yes! When I press my fingers right next to my breastbone it hurts sharply, and taking a deep breath feels like a stitch. No cold sweats, but my heart is pounding at 138 bpm."

■ **Arovia Health Assistant:** "While focal chest wall tenderness and deep inspiration pain point toward costochondritis, ongoing chest tightness with a resting heart rate of 138 bpm and dizziness in someone on oral contraceptives cannot be cleared remotely without an emergency ECG and cardiopulmonary evaluation (ruling out SVT, acute arrhythmia, or pulmonary embolism). Call 112 for emergency ambulance transport; do not attempt to drive yourself."

■ **Aanya Sen:** "Understood doctor, my roommate is calling the 112 ambulance right now. Resting quietly and breathing normally until they arrive."

■ **CLINICAL INFERENCE: Triage: URGENT (RULE-OUT) | Diagnosis: Costochondritis / Musculoskeletal Chest Wall Strain (Posterior: 64.4%, Multiplier: 2.8x)**
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

PATIENT CONSULTATION: #5 VIKRAMADITYA 'VIK' RATHORE (19y male) — Collegiate 400m Sprinter & Kinesiology Student

■ **Vikramaditya 'Vik' Rathore:** "Hey doc. During my 400m sprint intervals today, right at the 300m bend during max effort, I got this sudden sharp stitch across my left 4th and 5th costochondral joints. Exquisite focal tenderness on palpation. Resting HR is 46 bpm, SpO2 is 99%."

■ **Arovia Health Assistant:** "Hello Vikramaditya. To pinpoint this: does the pain become sharply worse when you shrug your shoulders or expand your ribcage fully, and does it radiate to your neck or arm?"

■ **Vikramaditya 'Vik' Rathore:** "Exactly. Deep inspiration spikes the pain to 7/10 right at the chondral junction, but if I breathe shallowly, it's barely a 2/10. No radiation to neck or arm at all."

■ **Arovia Health Assistant:** "Reproducible tenderness at the costochondral junction that spikes with deep inspiration strongly suggests Costochondritis from heavy training. However, exertional chest pain can never be cleared as zero cardiac risk remotely without an ECG. Rest from strenuous training, apply cool/warm compresses, and get a baseline screening ECG at your sports medicine clinic."

■ **Vikramaditya 'Vik' Rathore:** "Understood doc, skipping heavy upper body lifting and booking a screening ECG at our campus sports clinic today."

■ **CLINICAL INFERENCE: Triage: URGENT (RULE-OUT) | Diagnosis: Costochondritis / Musculoskeletal Chest Wall Strain (Posterior: 64.4%, Multiplier: 2.8x)**
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

2. Pulmonology: 103°F High Fever, Shaking Rigors & Green Phlegm

Clinical Domain: Infectious Disease / Pneumonia vs. Sepsis vs. Byssinosis | **Complaint:** High-grade spiking fever with uncontrollable shivering rigors, rust/yellow-green thick sputum, and pleuritic right chest stitch

A. Complete 18-Persona Risk & Organ Strain Comparison Matrix

Persona	Triage	Top Calculated Diagnosis	Posterior	Prior Shift (α)	Key Organ Strain	Primary DDI / Safety Alert
#1 Ramesh Sharma 68y M	EMERGENCY	**Community-Acquired Lobar Pneumonia**	**59.5%**	2.2x	Resp: 100% Cardio: 85%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#2 Aanya Sen 23y F	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#3 Gurpreet Singh 42y M	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 100% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#4 Meera Nambiar 32y F	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	■ **PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimest...**
#5 Vikramaditya 'Vik' Rathore 19y M	URGENT	**Community-Acquired Lobar Pneumonia**	**53.3%**	1.2x	Resp: 100% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#6 Devaki Patel 74y F	EMERGENCY	**Community-Acquired Lobar Pneumonia**	**59.5%**	2.2x	Resp: 95% Cardio: 70%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#7 Sunita Bannerjee 49y F	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 100% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#8 Karthik 'Koko' Roy 28y M	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#9 Dr. Farooq Qureshi 58y M	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 100% Cardio: 85%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#10 Priyanka Rao 36y F	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#11 Anand Joshi 54y M	EMERGENCY	**Community-Acquired Lobar Pneumonia**	**59.8%**	3.8x	Resp: 100% Cardio: 60%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#12 Zoya Sheikh 16y F	EMERGENCY	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#13 Harbans Lal 81y M	EMERGENCY	**Community-Acquired Lobar Pneumonia**	**59.5%**	2.2x	Resp: 100% Cardio: 95%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#14 Radhika Menon 41y F	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#15 Manish Chawla 47y M	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 75%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#16 Tenzin Norbu 37y M	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#17 Shalini Kulkarni 63y F	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 60%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#18 Deepak Verma 34y M	URGENT	**Community-Acquired Lobar Pneumonia**	**48.8%**	1.0x	Resp: 95% Cardio: 75%	✓ No immediate high-risk DDI clashes detected with baseline profile.

B. Synthetic Clinical Simulation Transcripts (Multi-Turn Triage Benchmark)

PATIENT CONSULTATION: #11 ANAND JOSHI (54y male) — Senior Chemical Dye & Textile Mill Foreman

■ **Anand Joshi:** "Doctor sahab, namaste. 30 saal se textile mill mein cotton dust mein kaam karta hoon, subah ki khansi aam baat hai. Lekin 4 din se 103 degree bukhar hai, kaanpte hue thand lag rahi hai, aur khansi mein peela-hara balgam aa raha hai. Saans lene mein gale mein ghutan ho rahi hai."

■ **Arovia Health Assistant:** "Hello Anand ji. Given your 30 years of mill dust exposure and smoking, your lung reserve is already vulnerable. Are you able to speak a full sentence without gasping for air?"

■ **Anand Joshi:** "Nahi doctor sahab... Do lafz bolte hi saans phool rahi hai. Seene ke daayein taraf sui jaisi chubhan ho rahi hai."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: Severe breathlessness with inability to complete sentences, high fever, and purulent sputum points to Community-Acquired Pneumonia with impending respiratory failure. Call 112 for ambulance transport to the hospital immediately."

■ **Anand Joshi:** "Haanji sahab, mera beta 112 ambulance bula raha hai abhi. Hospital ja rahe hain."

■ CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Community-Acquired Lobar Pneumonia (Posterior: 59.8%, Multiplier: 3.8x)
Safety / DDI Filter: ■ GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe gastric mucosal erosion and upper gastrointestinal hemorrhage risk.

PATIENT CONSULTATION: #13 HARBANS LAL (81y male) — Assisted Living Resident / Retired Railway Mechanical Engineer

■ **Harbans Lal:** "...Where is the station master? The 4:15 engine is whistling in the cold... (Nursing staff: 'Babuji has 103°F fever, coughing thick green sputum, and breathing rapidly at 28 breaths/min')."

■ **Arovia Health Assistant:** "Hello nursing team. In an 81-year-old with heart failure (EF 32%) and vascular dementia, high fever with tachypnea is high-risk Pneumonia with Sepsis concern. What are his current vitals and blood pressure?"

■ **Harbans Lal:** "...My legs are shivering under the railway blanket... (Staff: 'BP dropped to 94/56 mmHg, SpO2 is 87% on room air, pulse 112 weak, patient is severely confused')."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY — SEVERE SEPSIS: Apply supplemental oxygen immediately to support saturation. Transfer via Advanced Life Support (ALS) ambulance to the emergency department for arterial blood gas, blood cultures, and IV antibiotics."

■ **Harbans Lal:** "(Staff: 'ALS ambulance dispatched, high-flow oxygen mask placed, vitals chart ready for the emergency team.')"

■ CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Community-Acquired Lobar Pneumonia (Posterior: 59.5%, Multiplier: 2.2x)
Safety / DDI Filter: ■ RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly contraindicated in Stage 3-5 CKD. High risk of hemodynamically mediated Acute Kidney Injury (AKI) and accelerating CKD progression via afferent arteriolar vasoconstriction.

PATIENT CONSULTATION: #12 ZOYA SHEIKH (16y female) — High School Science Student & Competitive Swimmer

■ **Zoya Sheikh:** "Doctor, I thought I just caught a cold after swim training, but I've had a 103°F fever for 4 days with violent shaking rigors! I'm coughing up thick yellow-green phlegm, and my right side stabs when I inhale. My pulse oximeter shows 93% SpO2."

■ **Arovia Health Assistant:** "Hello Zoya. A 4-day 103°F fever with shaking rigors, green sputum, pleuritic chest pain, and SpO2 of 93% in a teenager indicates acute Community-Acquired Pneumonia. Are you breathless when walking or feeling dizzy?"

■ **Zoya Sheikh:** "I get breathless just walking to the bathroom, and my skin eczema is flaring from all the fever sweat. No wheezing, just deep rattling cough."

■ **Arovia Health Assistant:** "Because your oxygen saturation has dropped to 93% and you are breathless with minimal exertion, you need immediate in-person hospital evaluation. Please have your parents take you to the Emergency Room right now for a chest X-ray and clinical assessment."

■ **Zoya Sheikh:** "Okay doctor, my mother is driving me to the hospital emergency room right now."

■ CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Community-Acquired Lobar Pneumonia (Posterior: 48.8%, Multiplier: 1.0x)
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

3. Rheumatology: 3 AM Screaming Big Toe Swelling & Calor (Podagra)

Clinical Domain: Metabolic / Acute Gout vs. Pseudogout vs. Septic Joint | **Complaint:** Woke up at 3 AM with sensation of boiling water poured on right big toe. Joint is fiery red, swollen 2x size, exquisitely tender to touch

A. Complete 18-Persona Risk & Organ Strain Comparison Matrix

Persona	Triage	Top Calculated Diagnosis	Posterior	Prior Shift (α)	Key Organ Strain	Primary DDI / Safety Alert
#1 Ramesh Sharma 68y M	**URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**80.0%**	2.8x	Renal: 100% Cardio: 50%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#2 Aanya Sen 23y F	**URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**40.9%**	0.25x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#3 Gurpreet Singh 42y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**73.4%**	1.0x	Renal: 80% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#4 Meera Nambiar 32y F	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**49.2%**	0.35x	Renal: 65% Cardio: 25%	■ **PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimest...**
#5 Vikramaditya 'Vik' Rathore 19y M	**URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**40.9%**	0.25x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#6 Devaki Patel 74y F	**URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**61.0%**	1.0x	Renal: 80% Cardio: 35%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#7 Sunita Bannerjee 49y F	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**49.2%**	0.35x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#8 Karthik 'Koko' Roy 28y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**73.4%**	1.0x	Renal: 65% Cardio: 25%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#9 Dr. Farooq Qureshi 58y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**93.3%**	5.0x	Renal: 65% Cardio: 50%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#10 Priyanka Rao 36y F	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**49.2%**	0.35x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#11 Anand Joshi 54y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**73.4%**	1.0x	Renal: 65% Cardio: 25%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#12 Zoya Sheikh 16y F	**URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**40.9%**	0.25x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#13 Harbans Lal 81y M	**URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**81.4%**	2.8x	Renal: 100% Cardio: 60%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#14 Radhika Menon 41y F	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**49.2%**	0.35x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#15 Manish Chawla 47y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**96.8%**	11.0x	Renal: 65% Cardio: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#16 Tenzin Norbu 37y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**85.9%**	2.2x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#17 Shalini Kulkarni 63y F	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**73.4%**	1.0x	Renal: 65% Cardio: 25%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#18 Deepak Verma 34y M	**SEMI-URGENT**	**Acute Gouty Arthritis (Podagra Attack)**	**73.4%**	1.0x	Renal: 65% Cardio: 40%	✓ No immediate high-risk DDI clashes detected with baseline profile.

B. Synthetic Clinical Simulation Transcripts (Multi-Turn Triage Benchmark)

PATIENT CONSULTATION: #15 MANISH CHAWLA (47y male) — Commercial Real Estate Developer & Broker

■ **Manish Chawla:** "Doctor sahab! Save me! Last night I closed a 25-crore land deal, celebrated with tandoori mutton and 4 pegs of scotch. Woke up at 3 AM screaming—my right big toe is swollen like an apple and glowing fiery red! Even the ceiling fan air touching it feels like a blowtorch!"

■ **Arovia Health Assistant:** "Hello Mr. Chawla. Acute nocturnal podagra triggered by purine-heavy foods and alcohol is classical for Gout. Have you had fever or chills, and have you taken any medications yet?"

■ **Manish Chawla:** "No fever or chills, just pure joint fire! My uric acid was 9.8 mg/dL last month. I took two Brufen tablets an hour ago, but my BP cuff shows 156/96 and the pain is still 10/10!"

■ **Arovia Health Assistant:** "With BP 156/96 and repeated NSAID ingestion, avoid taking further Brufen. Apply a cool ice pack wrapped in a cloth for 15 minutes (avoid warm water which worsens swelling). Contact your doctor for a prescription of Colchicine or oral Prednisolone, and screen for infection."

■ **Manish Chawla:** "Understood doctor sahab. Putting ice pack on toe now, no more Brufen. Calling my physician for the Colchicine prescription."

■ **CLINICAL INFERENCE: Triage: SEMI-URGENT | Diagnosis: Acute Gouty Arthritis (Podagra Attack) (Posterior: 96.8%, Multiplier: 11.0x)**
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

PATIENT CONSULTATION: #1 RAMESH SHARMA (68y male) — Retired Bank Clerk

■ **Ramesh Sharma:** "Beta, since 4 AM my right big toe has swollen up red and burning hot. I thought of taking a Brufen tablet from my drawer to kill the pain so I could walk to the temple."

■ **Arovia Health Assistant:** "STOP! Do NOT take Brufen! Because you have Stage 3 Chronic Kidney Disease (eGFR 48), NSAIDs carry a severe hazard of precipitating acute kidney injury. Are you having fever, or any break in the skin near the toe?"

■ **Ramesh Sharma:** "No fever, no skin break, beta. Joint is fiery hot. Thank God I asked you before swallowing the Brufen! What should I do right now for relief?"

■ **Arovia Health Assistant:** "Use Paracetamol 650mg for mild relief, elevate your foot on two pillows, and apply a cool ice pack wrapped in a towel for 15 minutes. Avoid warm Epsom soaking. Consult your physician today for a renally adjusted anti-inflammatory."

■ **Ramesh Sharma:** "Understood beta, threw the Brufen strip away. I am elevating my foot with a cool pack now and will visit my doctor this morning."

■ **CLINICAL INFERENCE: Triage: URGENT | Diagnosis: Acute Gouty Arthritis (Podagra Attack) (Posterior: 80.0%, Multiplier: 2.8x)**
Safety / DDI Filter: ■ RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly contraindicated in Stage 3-5 CKD. High risk of hemodynamically mediated Acute Kidney Injury (AKI) and accelerating CKD progression via afferent arteriolar vasoconstriction.

PATIENT CONSULTATION: #9 DR. FAROOQ QURESHI (58y male) — Senior Cardiothoracic Surgeon

■ **Dr. Farooq Qureshi:** "Colleague, I just finished a 6-hour CABG scrub while slightly dehydrated, and woke up at 3 AM with acute podagra of the right 1st MTP joint. Erythematous, edematous, 9/10 allodynia. Given my DAPT (Aspirin + Clopidogrel) and Ramipril, I am holding off on NSAIDs."

■ **Arovia Health Assistant:** "Hello Dr. Qureshi. Volume depletion on an ACE inhibitor during prolonged standing is a classic precipitant. What was your most recent serum urate and baseline creatinine, and are you afebrile?"

■ **Dr. Farooq Qureshi:** "Afebrile. Last serum urate was 8.6 mg/dL, creatinine 1.1 mg/dL. Pain is strictly monoarticular in the right hallux. Zero systemic signs."

■ **Arovia Health Assistant:** "With active DAPT, systemic NSAIDs are strictly contraindicated due to severe GI hemorrhage and cardiovascular stent risk. A short course of oral Prednisolone or low-dose Colchicine (screening for additive statin-colchicine skeletal myopathy with your Rosuvastatin), plus cool compression and limb elevation, is the safest regimen."

■ **Dr. Farooq Qureshi:** "Good catch on the additive colchicine-statin myotoxicity and rhabdomyolysis hazard—I will opt for a short Prednisolone taper and hydrate. Thank you colleague."

■ **CLINICAL INFERENCE: Triage: SEMI-URGENT | Diagnosis: Acute Gouty Arthritis (Podagra Attack) (Posterior: 93.3%, Multiplier: 5.0x)**
Safety / DDI Filter: ■ HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High bleeding hazard. NSAIDs contraindicated (severe GI bleed risk); avoid unmonitored procedures; record last-dose timing for emergency team.

4. Gastroenterology: Coffee-Ground Vomitus, Melena & Epigastric Agony

Clinical Domain: Upper GI Hemorrhage / NSAID Peptic Ulcer vs. Varices | **Complaint:** Gnawing burn in upper stomach, sudden vomiting of dark coffee-ground speckled material, and passage of black tarry foul-smelling stools

A. Complete 18-Persona Risk & Organ Strain Comparison Matrix

Persona	Triage	Top Calculated Diagnosis	Posterior	Prior Shift (α)	Key Organ Strain	Primary DDI / Safety Alert
#1 Ramesh Sharma 68y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**77.8%**	2.8x	GI: 98% Cardio: 100%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#2 Aanya Sen 23y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#3 Gurpreet Singh 42y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**81.4%**	3.5x	GI: 100% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#4 Meera Nambiar 32y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**55.6%**	1.0x	GI: 98% Cardio: 75%	■ **PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimest...**
#5 Vikramaditya 'Vik' Rathore 19y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#6 Devaki Patel 74y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 85%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#7 Sunita Bannerjee 49y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#8 Karthik 'Koko' Roy 28y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**69.2%**	1.8x	GI: 100% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#9 Dr. Farooq Qureshi 58y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**82.0%**	2.5x	GI: 98% Cardio: 100%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#10 Priyanka Rao 36y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#11 Anand Joshi 54y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**76.6%**	1.8x	GI: 100% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#12 Zoya Sheikh 16y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#13 Harbans Lal 81y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**82.0%**	2.5x	GI: 98% Cardio: 100%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#14 Radhika Menon 41y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#15 Manish Chawla 47y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 100% Cardio: 90%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#16 Tenzin Norbu 37y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#17 Shalini Kulkarni 63y F	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 75%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#18 Deepak Verma 34y M	EMERGENCY	**Bleeding Peptic Ulcer (Gastric / Duodenal)**	**64.5%**	1.0x	GI: 98% Cardio: 90%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**

B. Synthetic Clinical Simulation Transcripts (Multi-Turn Triage Benchmark)

PATIENT CONSULTATION: #3 GURPREET SINGH (42y male) — Interstate Long-Haul Truck Driver

■ **Gurpreet Singh:** "Ustaad ji, namaste. I have been taking 600mg Brufen tablets twice a day for highway back pain. But today my stomach started burning like battery acid, and when I puked at the petrol pump, it looked like dark coffee grounds! And my stool is completely black like tar!"

■ **Arovia Health Assistant:** "Gurpreet, sit down immediately! Vomiting dark coffee grounds and passing black tarry stools means you have an acute Upper GI Bleed (likely an NSAID-induced ulcer). Are you feeling dizzy or faint?"

■ **Gurpreet Singh:** "Yes paaji, when I climbed down from the truck, my eyes went black and I almost collapsed on the gravel. Hands feel cold and clammy."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: You are in early hypovolemic shock from blood loss. Do not eat, drink, or take any medication. Have highway patrol or an ambulance transport you to the nearest hospital emergency department immediately."

■ **Gurpreet Singh:** "Understood ustaad ji. Calling highway ambulance 112 right now. Lying down on the truck bunk."

■ CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Bleeding Peptic Ulcer (Gastric / Duodenal) (Posterior: 81.4%, Multiplier: 3.5x)
Safety / DDI Filter: ■ GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe gastric mucosal erosion and upper gastrointestinal hemorrhage risk.

PATIENT CONSULTATION: #9 DR. FAROOQ QURESHI (58y male) — Senior Cardiothoracic Surgeon

■ **Dr. Farooq Qureshi:** "Colleague, I need urgent triage coordination. Over the past 6 hours I've had gnawing epigastric pain, and 20 minutes ago I had coffee-ground emesis followed by a large melenic stool. I am on daily Aspirin 75mg and Clopidogrel 75mg, and foolishly took Diclofenac for shoulder strain."

■ **Arovia Health Assistant:** "Dr. Qureshi, lie down immediately with your head elevated 30 degrees (semi-Fowler position) or in lateral recovery position to protect your airway and prevent aspiration while vomiting. Combining an NSAID with dual antiplatelet therapy has precipitated an acute Upper GI Hemorrhage. What is your current heart rate, blood pressure, and orthostatic status?"

■ **Dr. Farooq Qureshi:** "HR is 112 bpm, BP on my home cuff is 104/68 mmHg—down from my 142/90 baseline. Significant postural lightheadedness. Shock index > 1.0."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: Shock index > 1.0 indicates hemodynamic compromise. Do NOT independently stop or restart prescribed antiplatelets; immediately convey your exact medication chart (Aspirin 75mg, Clopidogrel 75mg, Diclofenac) to emergency and cardiology teams for specialist-coordinated hemostasis per NICE CG141 guidelines. Remain strictly NPO, and transfer via emergency cardiac ambulance to the ER for large-bore IV access, IV PPI infusion, type-and-cross, and emergency therapeutic endoscopy."

■ **Dr. Farooq Qureshi:** "Cardiac ambulance called, GI on-call notified. Resting in semi-Fowler position with head elevated 30 degrees. Thank you colleague."

■ CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Bleeding Peptic Ulcer (Gastric / Duodenal) (Posterior: 82.0%, Multiplier: 2.5x)
Safety / DDI Filter: ■ HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High bleeding hazard. NSAIDs contraindicated (severe GI bleed risk); avoid unmonitored procedures; record last-dose timing for emergency team.

PATIENT CONSULTATION: #13 HARBANS LAL (81y male) — Assisted Living Resident / Retired Railway Mechanical Engineer

■ **Harbans Lal:** "...Why did they spill black coal tar on the engine floor?... My stomach is burning... (Assisted Living Nurse: 'Doctor, Babuji just vomited dark brown coffee-ground liquid and passed black tarry stool in his diaper! He is on blood thinners for his heart!')."

■ **Arovia Health Assistant:** "Emergency alert for nursing staff: Coffee-ground vomitus and melena in an 81-year-old on anticoagulants with CHF is a life-threatening Upper GI Hemorrhage. What are his current vitals and level of consciousness?"

■ **Harbans Lal:** "...The boiler pressure is dropping, bring the coal... (Nurse: 'He is pale and clammy, pulse is 118 weak, BP dropped to 88/54 mmHg!')."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY — HYPOVOLEMIC SHOCK: Position him supine with lateral recovery tilt in case of emesis, protect his airway, and transfer via ALS ambulance immediately. Do NOT give oral fluids. Send his anticoagulant medication chart for urgent reversal protocol."

■ **Harbans Lal:** "(Nurse: 'Turned on side, oxygen mask placed, ALS ambulance pulling into the facility gate right now.')"

■ CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Bleeding Peptic Ulcer (Gastric / Duodenal) (Posterior: 82.0%, Multiplier: 2.5x)
Safety / DDI Filter: ■ HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High bleeding hazard. NSAIDs contraindicated (severe GI bleed risk); avoid unmonitored procedures; record last-dose timing for emergency team.

5. Nephrology: Watery Diarrhea (12x/day), Sunken Eyes & Dehydration

Clinical Domain: Pre-Renal AKI vs. Secretory Enteritis & Metformin Lactic Acidosis | **Complaint:** Continuous rice-water or watery diarrhea stools every 45 mins, severe abdominal cramping, intense dry mouth, dark scanty urine

A. Complete 18-Persona Risk & Organ Strain Comparison Matrix

Persona	Triage	Top Calculated Diagnosis	Posterior	Prior Shift (α)	Key Organ Strain	Primary DDI / Safety Alert
#1 Ramesh Sharma 68y M	EMERGENCY	**Pre-Renal Acute Kidney Injury (Hypovolemic AKI)**	**71.8%**	8.4x	Renal: 100% GI: 90%	■■ **METFORMIN SICK-DAY RULE (SADMANS): Hold Metformin during dehydrating illness, a...**
#2 Aanya Sen 23y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#3 Gurpreet Singh 42y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 100% GI: 100%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#4 Meera Nambiar 32y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	■ **PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimest...**
#5 Vikramaditya 'Vik' Rathore 19y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#6 Devaki Patel 74y F	EMERGENCY	**Pre-Renal Acute Kidney Injury (Hypovolemic AKI)**	**43.1%**	2.5x	Renal: 100% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#7 Sunita Bannerjee 49y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#8 Karthik 'Koko' Roy 28y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 100%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#9 Dr. Farooq Qureshi 58y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#10 Priyanka Rao 36y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#11 Anand Joshi 54y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 100%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#12 Zoya Sheikh 16y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#13 Harbans Lal 81y M	EMERGENCY	**Pre-Renal Acute Kidney Injury (Hypovolemic AKI)**	**76.1%**	10.5x	Renal: 100% GI: 90%	■ **RENAL NSAID CONTRAINDICATION: NSAIDs (Ibuprofen/Naproxen/Diclofenac) strictly co...**
#14 Radhika Menon 41y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#15 Manish Chawla 47y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 100%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#16 Tenzin Norbu 37y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#17 Shalini Kulkarni 63y F	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#18 Deepak Verma 34y M	URGENT	**Acute Secretory Gastroenteritis with Severe Dehydration**	**61.5%**	1.0x	Renal: 95% GI: 90%	✓ No immediate high-risk DDI clashes detected with baseline profile.

B. Synthetic Clinical Simulation Transcripts (Multi-Turn Triage Benchmark)

PATIENT CONSULTATION: #1 RAMESH SHARMA (68y male) — Retired Bank Clerk

■ **Ramesh Sharma:** "Beta, since yesterday morning I have had 11 or 12 watery loose motions. My mouth is like paper, and I took my morning Metformin and BP tablet as usual with a sip of water."

■ **Arovia Health Assistant:** "CRITICAL SICK-DAY RULE: Stop taking your Metformin immediately! In severe dehydration, Metformin accumulation carries an increased risk of serious Lactic Acidosis (MALA). How is your urination and dizziness?"

■ **Ramesh Sharma:** "Beta, I haven't passed any urine since 8 hours. When I stand up, the room spins and I have to grab the wall. My calves are cramping painfully."

■ **Arovia Health Assistant:** "Your symptoms indicate acute pre-renal dehydration risk with anuria. Sip WHO Oral Rehydration Salts (ORS) slowly every 10 minutes, and have your family take you to an emergency clinic immediately for IV fluid evaluation."

■ **Ramesh Sharma:** "Understood beta, pausing Metformin and starting ORS right away. My son is taking me to the emergency clinic now."

■ **CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Pre-Renal Acute Kidney Injury (Hypovolemic AKI) (Posterior: 71.8%, Multiplier: 8.4x)**
Safety / DDI Filter: ■■ METFORMIN SICK-DAY RULE (SADMANS): Hold Metformin during dehydrating illness, acute sepsis, or volume depletion due to increased risk of serious Metformin-Associated Lactic Acidosis (MALA). Coordinate clinical restart with physician once eGFR and hydration normalize.

PATIENT CONSULTATION: #10 PRIYANKA RAO (36y female) — Permaculture Farmer & Ashtanga Yoga Acharya

■ **Priyanka Rao:** "Namaste doctor. My digestive fire (Agni) is severely disturbed. Since yesterday, I have had 12 profuse watery evacuations, abdominal cramping, and extreme dryness. I prefer natural healing methods—can I just drink coconut water?"

■ **Arovia Health Assistant:** "Namaste Priyanka. While I respect your holistic philosophy, losing fluid 12 times in 24 hours has placed your kidneys under severe strain. Plain coconut water is rich in potassium but lacks the equimolar sodium-glucose ratio (SGLT1) needed to stop secretory dehydration. Are you dizzy on standing?"

■ **Priyanka Rao:** "Yes doctor, my urine is dark amber and scanty, and I nearly blacked out when standing up from my yoga mat."

■ **Arovia Health Assistant:** "CRITICAL DEHYDRATION HAZARD: Because you are nearly blacking out on standing, you MUST drink WHO-formula Oral Rehydration Salts (or emergency home ORS: 1L boiled water + 6 level tsp sugar + 1/2 tsp salt) at 200ml after every stool, and proceed to an urgent care clinic immediately for IV fluid assessment."

■ **Priyanka Rao:** "Thank you for explaining the physiology clearly, doctor. My husband is mixing the ORS now and driving me to the local clinic."

■ **CLINICAL INFERENCE: Triage: URGENT | Diagnosis: Acute Secretory Gastroenteritis with Severe Dehydration (Posterior: 61.5%, Multiplier: 1.0x)**
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

PATIENT CONSULTATION: #4 MEERA NAMBIAR (32y female) — High School Biology Teacher

■ **Meera Nambiar:** "Doctor, I'm in my second trimester of pregnancy and I've had over 10 profuse watery diarrhea stools today. My mouth is completely parched, my urine is very dark, and I feel dizzy sitting up."

■ **Arovia Health Assistant:** "Hello Meera. In pregnancy, rapid dehydration reduces uteroplacental perfusion and can stimulate premature uterine contractions. Are you feeling normal fetal movements, and are you having any lower abdominal cramping or tightening?"

■ **Meera Nambiar:** "I can sip water, and the baby kicked an hour ago, but my lower belly feels crampy and my heart is racing."

■ **Arovia Health Assistant:** "EMERGENCY OBSTETRIC PRECAUTION: With gestational anemia (Hb 9.8), tachycardia, and dehydration, oral fluids at home are not enough. Start sipping WHO-ORS continuously and report to Maternity Triage / Labor & Delivery immediately for IV fluid rehydration and fetal monitoring."

■ **Meera Nambiar:** "Sipping ORS now and heading straight to Maternity Triage with my husband. Thank you, doctor."

■ **CLINICAL INFERENCE: Triage: URGENT | Diagnosis: Acute Secretory Gastroenteritis with Severe Dehydration (Posterior: 61.5%, Multiplier: 1.0x)**
Safety / DDI Filter: ■ PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimester (premature ductus arteriosus closure, oligohydramnios). ACE inhibitors, ARBs, and Fluoroquinolones contraindicated due to fetotoxicity. Paracetamol is the first-line analgesic of choice when clinically required.

6. Neurology: Acute FAST Stroke (Facial Droop, Left Arm Weakness)

Clinical Domain: Cerebrovascular / Ischemic Infarction vs. Hemorrhage vs. TIA | **Complaint:** Sudden onset inability to move left arm, mouth drooping on left side when smiling, slurred words, unable to repeat a simple sentence

A. Complete 18-Persona Risk & Organ Strain Comparison Matrix

Persona	Triage	Top Calculated Diagnosis	Posterior	Prior Shift (α)	Key Organ Strain	Primary DDI / Safety Alert
#1 Ramesh Sharma 68y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**83.4%**	2.8x	Neuro: 98% Cardio: 90%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#2 Aanya Sen 23y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**36.8%**	0.2x	Neuro: 100% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#3 Gurpreet Singh 42y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**79.8%**	2.2x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#4 Meera Nambiar 32y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**74.5%**	1.0x	Neuro: 98% Cardio: 65%	■ **PREGNANCY MEDICATION SAFETY (FDA PLLR): Avoid systemic NSAIDs in 2nd/3rd trimest...**
#5 Vikramaditya 'Vik' Rathore 19y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**36.8%**	0.2x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#6 Devaki Patel 74y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**92.9%**	4.48x	Neuro: 100% Cardio: 75%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#7 Sunita Bannerjee 49y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**86.5%**	2.2x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#8 Karthik 'Koko' Roy 28y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**74.5%**	1.0x	Neuro: 100% Cardio: 65%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#9 Dr. Farooq Qureshi 58y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**91.1%**	3.52x	Neuro: 98% Cardio: 90%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#10 Priyanka Rao 36y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**74.5%**	1.0x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#11 Anand Joshi 54y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**86.5%**	2.2x	Neuro: 98% Cardio: 65%	■ **GASTROINTESTINAL CONTRAINDICATION: Systemic NSAIDs contraindicated due to severe...**
#12 Zoya Sheikh 16y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**36.8%**	0.2x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#13 Harbans Lal 81y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**88.9%**	4.48x	Neuro: 100% Cardio: 100%	■ **HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High ble...**
#14 Radhika Menon 41y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**74.5%**	1.0x	Neuro: 100% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#15 Manish Chawla 47y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**79.8%**	2.2x	Neuro: 98% Cardio: 80%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#16 Tenzin Norbu 37y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**74.5%**	1.0x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#17 Shalini Kulkarni 63y F	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**74.5%**	1.0x	Neuro: 98% Cardio: 65%	✓ No immediate high-risk DDI clashes detected with baseline profile.
#18 Deepak Verma 34y M	EMERGENCY	**Acute Ischemic Stroke (Cerebral Infarction)**	**86.5%**	2.2x	Neuro: 98% Cardio: 80%	✓ No immediate high-risk DDI clashes detected with baseline profile.

B. Synthetic Clinical Simulation Transcripts (Multi-Turn Triage Benchmark)

PATIENT CONSULTATION: #13 HARBANS LAL (81y male) — Assisted Living Resident / Retired Railway Mechanical Engineer

■ **Harbans Lal:** "...The railway signals are crossed... I can't lift my left hand to touch my tea cup... (Assisted Living Nurse: 'Doctor, Babuji's left mouth corner suddenly started drooping 25 minutes ago during breakfast, and his left arm is completely limp!')."

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: Babuji is exhibiting classic Acute Stroke signs (FAST Positive). Do NOT give any food, water, or aspirin. Is an emergency ambulance called, and what blood thinners is he currently taking?"

■ **Harbans Lal:** "(Nurse: '112 ambulance is dialed and arriving in 5 minutes! He takes Apixaban 2.5mg BID for atrial fibrillation. We have logged exact symptom onset as 8:15 AM.')."

■ **Arovia Health Assistant:** "Keep him positioned with head supported at 30 degrees. Hand the paramedics his medication chart showing Apixaban so the stroke team can evaluate whether mechanical thrombectomy is indicated over IV thrombolysis."

■ **Harbans Lal:** "(Nurse: 'Paramedics just wheeled the stretcher into his room. Handing over the Apixaban chart and timestamp now. Thank you!')"

■ **CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Acute Ischemic Stroke (Cerebral Infarction) (Posterior: 88.9%, Multiplier: 4.48x)**
Safety / DDI Filter: ■ HEMOSTATIC ALERT: Patient on active antiplatelet/anticoagulant therapy. High bleeding hazard. NSAIDs contraindicated (severe GI bleed risk); avoid unmonitored procedures; record last-dose timing for emergency team.

PATIENT CONSULTATION: #14 RADHIKA MENON (41y female) — Clinical Psychologist & Psychotherapist

■ **Radhika Menon:** "Doctor, I was in the middle of a therapy session with a client 20 minutes ago when my speech suddenly became slurred and thick. When I looked in my office mirror, the left corner of my mouth was drooping, and I just dropped my notepad because my left hand has zero grip strength! Is this a stroke?!"

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: Radhika, sudden unilateral facial droop, motor paralysis, and dysarthria are classic signs of an Acute Stroke (FAST Positive). Do NOT take food, water, or aspirin until a CT scan rules out hemorrhage. Is someone with you right now?"

■ **Radhika Menon:** "Yes, the clinic receptionist is in my room right now and dialing 112... my whole left arm feels completely dead..."

■ **Arovia Health Assistant:** "Have the receptionist record the exact symptom onset time (20 minutes ago). Lie down with your head elevated at 30 degrees. Transferring to a Stroke-Ready Hospital immediately is critical for acute evaluation."

■ **Radhika Menon:** "(Receptionist: 'Ambulance is 4 minutes away, onset time logged at 10:10 AM, keeping her calm and elevated. Thank you!')"

■ **CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Acute Ischemic Stroke (Cerebral Infarction) (Posterior: 74.5%, Multiplier: 1.0x)**
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

PATIENT CONSULTATION: #18 DEEPAK VERMA (34y male) — High-Frequency Financial Derivatives Trader

■ **Deepak Verma:** "Doc, help—I'm at my trading desk and suddenly couldn't type on the left side of my keyboard! My left arm won't lift, my co-trader says the left side of my face is sagging, and my words are coming out garbled! My BP has been running 144/92 for months and I skipped my meds today!"

■ **Arovia Health Assistant:** "CRITICAL EMERGENCY: Deepak, sudden left-sided arm weakness, facial droop, and slurred speech in the setting of uncontrolled hypertension is an Acute Stroke until proven otherwise. Do NOT swallow aspirin or BP pills right now. Has an ambulance been called?"

■ **Deepak Verma:** "(Co-trader typing: 'Yes, building security called 112, paramedics are on the way up the elevator. His left arm is totally limp and he has a pounding headache.')."

■ **Arovia Health Assistant:** "Keep him resting with his head slightly elevated, loosen his collar/tie, and give nothing by mouth. Inform the paramedics of the exact onset time and his untreated hypertension so the ER can rush him straight to the CT scanner."

■ **Deepak Verma:** "(Co-trader: 'Paramedics just walked onto the trading floor with the stretcher. Got the exact timestamp logged. Thanks doc.')"

■ **CLINICAL INFERENCE: Triage: EMERGENCY | Diagnosis: Acute Ischemic Stroke (Cerebral Infarction) (Posterior: 86.5%, Multiplier: 2.2x)**
Safety / DDI Filter: ✓ No immediate high-risk DDI clashes detected with baseline profile.

Physician Verification, Audit & Concurrence Protocol

This section is reserved for the attending medical doctor to review the mathematical calculations, cross-persona disease responses, and drug safety contraindications generated by the deterministic local engine (Build 2026.10-LocalEngine).

CLINICAL AUDIT & REGULATORY BOUNDARY NOTICE:
This dossier and its Bayesian calculations are synthetic simulation benchmarks generated solely by Arovia's deterministic local engine (Build 2026.10-LocalEngine, Zero External API Dependency) for physician verification and engineering review. It is **NOT approved for unmonitored patient self-care** or autonomous clinical decision-making without licensed clinical supervision.

Physician Audit Domain	Verification Standard	Concurrence	Physician Notes
1. Bayesian Covariate Priors (α)	Covariate multipliers shift baseline odds based on age, CAD history, diabetes, and smoking.	[ ] Concur [ ] Modify [ ] Disagree	_____ _____
2. NSAID & Renal Safety Guard	Absolute contraindication of systemic NSAIDs in CKD Stages 3-5, active peptic ulcer/GI bleed, and 2nd/3rd trimester pregnancy.	[ ] Concur [ ] Modify [ ] Disagree	_____ _____
3. Metformin Sick-Day Protocol	Mandatory cessation of Metformin during dehydration/diarrhea to mitigate risk of Metformin-Associated Lactic Acidosis (MALA).	[ ] Concur [ ] Modify [ ] Disagree	_____ _____
4. Organ Strain & Frailty Index	0-100% organ strain index mapping (Cardio, Renal, Resp, GI, Neuro) and acute frailty surge.	[ ] Concur [ ] Modify [ ] Disagree	_____ _____
5. Multi-Modal Integrative Safety	Traditional herbal preparations screened for herb-drug interactions and organ safety.	[ ] Concur [ ] Modify [ ] Disagree	_____ _____

Reviewing Doctor Name: _____

Medical Council Registration No: _____

Hospital / Clinic: _____

Specialty: [ ] Internal Med [ ] Cardiology [ ] GP

Physician Signature: _____

Date: _____

Official Clinic Stamp:

_____

Clinical Approval Status:
[ ] Verified for Patient Guidance
[ ] Adjusted per Notes
[ ] Retest with Additional Diagnostics